

Deafness and ATs for Hearing

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

<https://cmu-intro-to-accessibility.github.io>

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Administrivia

- P3 is due in 2 weeks: Friday Oct 9, 11:59pm.
- Please reach out to your partner organizations to set up your first meeting!
- On Wednesday, we will cover Human Subjects Research and how we do ethical research. Useful for P3.

Sign up for Project Checkins

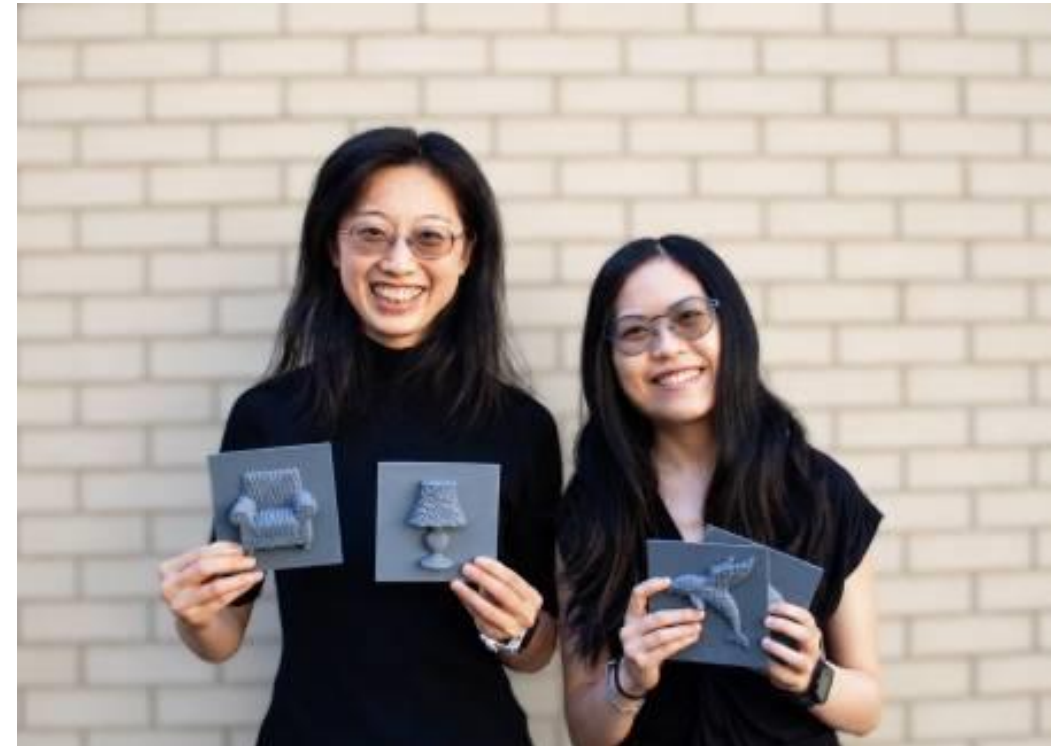
- Neeta and I will hold 30-minute project checkins.
 - P3 Checkin: October 5-9 (next week!)
 - P4 Checkin: October 19-23
 - P5 Checkin: November 2-6
 - P6 Checkin: November 16-20
- As a team, [sign up](https://bit.ly/cmu-project-checkin) for one (1) time slot per milestone.



<https://bit.ly/cmu-project-checkin>

Today's Accessibility Lunch

- **Ruihan Gao**, PhD student at CMU Robotics Institute
- Text-based Tactile Graphics Generation for the Visually Impaired
- 12:30pm in NSH 3305



Hearing Loss

- 34 million children in the world have disabling hearing loss.
 - 90% born to non-disabled, hearing parents.
- Routine hearing loss screening in USA was recommended in 1993. Was only adopted in 2005.
- Peak brain plasticity for language is <12 months of age.
- Can hearing loss be corrected?

Cochlear Implants



Hearing Aid

- Involve a microphone and “speaker” to sense environmental sounds (typically speech) and amplify the sound at appropriate frequencies.
- Hearing loss is not often experienced consistently across the speech frequency band. For instance, in aging, higher frequencies may be more impacted.
- The “speaker” is typically based on air waves or bone conduction.

Cochlear Implants and Hearing Aids

- 80% of children with hearing loss who got cochlear implants before 12 months of age achieved normal vocabulary development.
 - Dettman SJ, Dowell RC, Choo D, Arnott W, Abrahams Y, Davis A, Dornan D, Leigh J, Constantinescu G, Cowan R, Briggs RJ. Long-term Communication Outcomes for Children Receiving Cochlear Implants Younger Than 12 Months: A Multicenter Study. Otol Neurotol. 2016 Feb;37(2):e82-95. doi: 10.1097/MAO.0000000000000915. PMID: 26756160.
- Hearing aids provide comparable benefit to cochlear implants when given before 6 months old.
 - Fitzpatrick EM, Olds J, Gaboury I, McCrae R, Schramm D, Durieux-Smith A. Comparison of outcomes in children with hearing aids and cochlear implants. Cochlear Implants Int. 2012 Feb;13(1):5-15. doi: 10.1179/146701011X12950038111611. PMID: 22340747.

Cochlear Implants Are Transformative



Cochlear Implants Are Controversial...

Cochlear Implants Do Not *Fix* Hearing



“Deaf people like being Deaf, want to be Deaf, and are proud of their Deafness.”

- Deaf people share a language, a history, schools, sports, clubs, art, and movies.
 - Deaf people primarily socialize within their community.
 - 95% of Deaf marriages involve 2 deaf partners.
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- Barnett S. Clinical and cultural issues in caring for deaf people. Fam Med. 1999 Jan;31(1):17-22. PMID: 9987607.

Protecting and Interpreting Deaf Culture



Participation Activity

- What does the term hearing loss mean to you? How does it make you feel?
- Write an answer in **#in-class**:
 1. As a parent of a deaf newborn baby

Participation Activity

- What does the term hearing loss mean to you? How does it make you feel?
- Write an answer in **#in-class**:
 1. As a college student born deaf

Not hearing loss, deaf gain



Participation Activity

- What does the term hearing loss mean to you? How does it make you feel?
- Write two answers in **#in-class**:
 1. As someone who lost their hearing at age 18
 2. As someone who lost their hearing at age 78

Assistive Technologies Can Expose Social Complexity

- Is an assistive technology available to mitigate an impairment? Should it be?
- Can it *cure* the impairment or just reduce its impact?
- Who gets to decide whether an AT is used?
 - Self
 - Family
 - Social and Medical Peer Pressure
- The irreversible deployment of ATs for babies can bring people together at the same time as it destroys whole cultures.

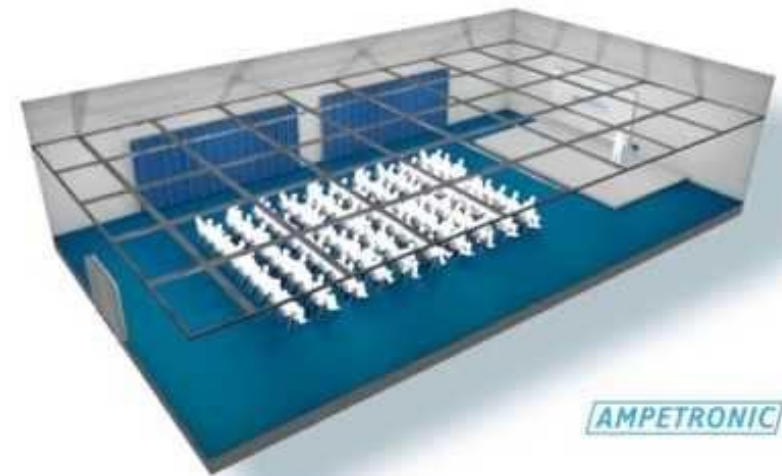
Assistive Technologies for Hearing Impairments

Assistive Listening Devices

Personal Listening Devices



Hearing Loop/Induction Loop



Captioning

Probably familiar with closed captioning for Television and Movies

Real-Time Captioning can be provided in multiple ways

- Computer-Assisted Remote Transcription (CART)
- Automatic Speech Recognition



The screenshot shows a video player interface. On the right is a video frame depicting a warrior in a Spartan helmet and armor, with the word "300" written in red over the image. On the left is a white text box with the following content:

Aside: Religious Viewpoint

Many pop culture references support this early belief system

Ephialtes in the 2006 version of 300

Many fictional and exaggerated, though they highlight some clear perspectives

Actually quite difficult to choose a side in this movie...

At the bottom left of the text box is a red circular logo with the letters "HCU" in white. At the bottom of the video frame, there is a black bar with white text that reads: "long ago but in this movie there's a character named ephialtes who was born with some physical matiz and in the culture of Sparta the warrior culture only people who". The words "ephaltes" and "matiz" are highlighted with red boxes.

TTY - Teletypewriter and Telephone Relay Services

Early solution telephone-based communication

Text based communication



Video Conferencing

- 2-way video call as early as 1930
- 1990s first desktop video conferencing
- 2000s begin smartphone-based video calls



Speaker Highlighting

- In Speaker View, video conferencing systems highlight the current speaker.
- What happens if the current speaker uses ASL?
- There are two flaws if there's an ASL interpreter in the meeting. What are they?
 1. Systems highlight the ASL interpreter instead of the ASL speaker.
 2. For the ASL speaker, systems switch to show oral speaker instead of the ASL interpreter!

Speech Recognition

- First speech recognition system from Bell Labs in 1952.
 - “Audrey” could recognize spoken numbers.
- Based on acoustics, linguistics: could recognize 1000 words by 1970s.
- Big leap in 1980s with Hidden Markov Models.
 - “Every time I fire a linguist, the performance of the system goes up.” - Fred Jelinek (IBM)
- Massive improvement in mid 2010s with deep learning
 - Word error rate dropped below 5% (better than people!)

AI-based ASL Recognition

- System that translates between sign language and English.
- Lots of research systems that can recognize some nouns from some ways of signing.
- However, no viable systems exist, but some startups are really trying:
 - [OmniBridge](#), [Rylo](#)
- Deaf community: Please do not attempt this without us. You're being audist.